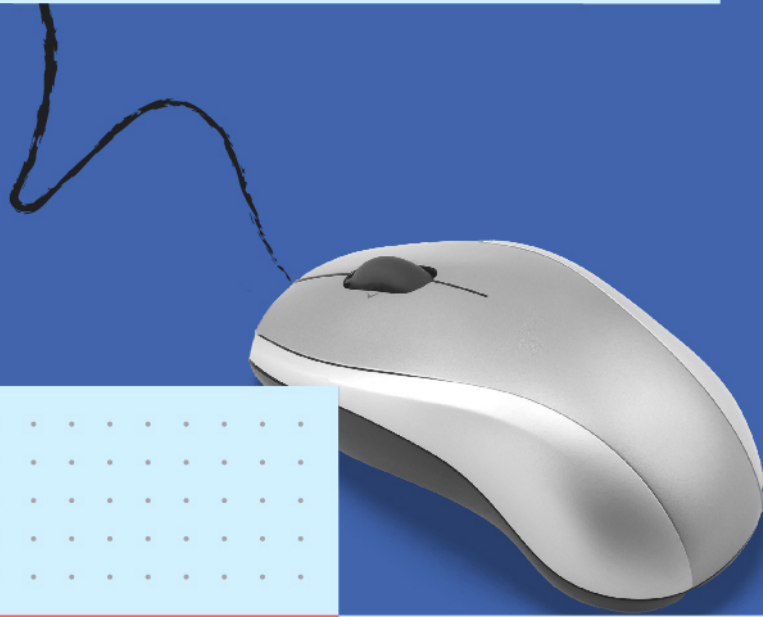




Caseware IDEA 12

CI202

IDEA Data Analysis Workbook

CI202

IDEA Data Analysis Workbook

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IDEA Academic Partnership
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A Caseware IDEA Document

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CI202 IDEA Data Analysis Workbook

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Preface

Welcome to the IDEA Workbook. IDEA is a comprehensive file interrogation tool for auditors, accountants, and investigators. With IDEA, you can:

- Analyze complex data
- Extract unusual items
- Create audit samples
- Identify duplicate transactions
- Carry out many other tasks that are easy to perform using the specially designed, intuitive Windows interface

This Workbook is designed for new IDEA users, and provides a more in-depth introduction to the application than the *IDEA Tutorial* that is provided with IDEA. The IDEA Workbook illustrates some common audit tests, with step-by-step instructions and solutions to explain how to perform three separate audits / investigations:

- Accounts Receivable Audit
- Accounts Payable Audit and Fraud Investigation
- Inventory Analysis

Each section within an audit / investigation has an audit objective as well as an IDEA lesson objective explaining the IDEA features used in the exercise.

The data files for each audit / investigation are provided as an accompanying download. The data files have been created in a variety of file formats to provide experience in importing data from different sources. The numbers of records and fields within each data file have been restricted for speed and practicality.

We hope you enjoy using IDEA.

We wish to acknowledge the assistance provided by Efrim Boritz and Malik Datardina and the University of Waterloo Centre for Information System Assurance (<http://accounting.uwaterloo.ca/uwcisa/>).

Section 1



Introduction to the IDEA Workbook

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1.1 Setting the Scene

For the purposes of the exercises in this book, you are an audit manager with a firm of chartered accountants based in Ontario, Canada.

Your client portfolio includes several medium-sized companies, family-owned businesses and retailers. To prosper in the ever-increasing competitive market, you have identified the need to:

- Improve the efficiency of audits
- Improve the quality of audits and business advice given to clients
- Provide added-value services to clients
- Win new clients by offering exciting new services

You have seen demonstrations and read articles on how the IDEA Data Analysis Software will enable you to meet these needs, and you have recently purchased IDEA. You have selected one of your premiere clients to pilot your use of IDEA. You are concerned that the levels of controls have not increased to meet the growth in the client's business, and after consultation with your client, you decide to use IDEA on the Accounts Receivable area of the audit. Additionally, the client asks that you investigate a suspected case of fraud in the Accounts Payable department, and perform an Inventory Analysis to assist with a problem they have identifying obsolete stock and the subsequent calculation of inventory provisions. All audits and investigations were highly successful and identified problems and weaknesses, which would have been extremely difficult to accomplish without the use of IDEA.

1.2 How This Workbook is Organized

This Workbook provides an overview of the process of auditing using IDEA and provides step-by-step instructions for carrying out the following audits / investigations that you conduct:

- Accounts Receivable Audit
- Accounts Payable Audit and Fraud Investigation
- Inventory Analysis

Each audit / investigation is a separate section of the Workbook. An overview of each section explains the objectives of the audit / investigation, how to plan for the testing, and how to request the required data.

A wide range of data file formats is provided with the Workbook via digital download so that you can experience importing different file types.

Instructions are provided for preparing for the exercises, including copying the relevant data files.

The exercises are designed to cover most tests and options within IDEA. Both the audit objectives and the exercise details are stated for each test carried out. It is advised that the exercises are completed in order, as many of the exercises depend on earlier exercises. However, you may wish

to repeat or review sections, particularly before conducting a similar live audit. Solutions are provided for each exercise and many images of menu or toolbar button selections, completed dialog boxes, resultant databases, and reports are included for clarity.



You can customize many options, including toolbars, within IDEA. This Workbook provides instructions and images using the default toolbars and settings.

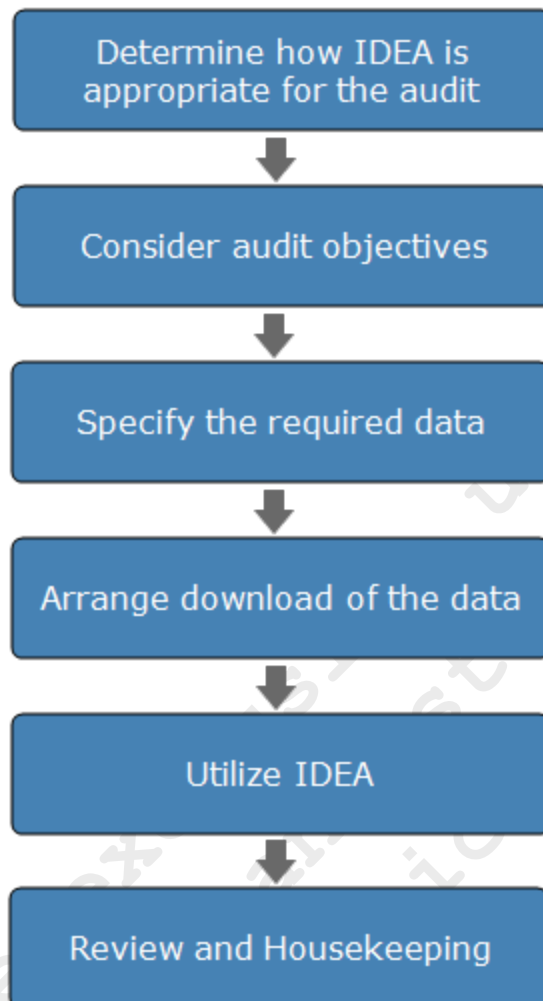
It is recommended to read the following section that outlines the use of IDEA before starting the exercises.

1.3 Overview of the Use of IDEA

1.3.1 Stages of Using IDEA

The exact process of using IDEA varies with circumstances, but the main steps are set out in the following chart.

The first three stages are planning stages and are key to getting the most out of IDEA.



1.3.2 Determine How IDEA is Best Utilized

To decide how to use IDEA on an audit, a brief evaluation of the following topics should be completed with an assessment of whether the benefits outweigh the costs:

- Key audit objectives and problems
- Volume and depth of information held on computer
- Ease of downloading data

Objectives and Problems

IDEA is most useful when there is a problem to solve on an audit. It may be assessing the extent of bad loans, debts, or inventory provisions, looking at performance in collecting cash or simply speeding up the time to prove calculations. Whatever the reason, there should be a core purpose which drives the use and benefits of IDEA.

IDEA is generally used for substantive testing. When there is a need to determine if items are in error, a problem needs to be quantified, or certain items identified, then IDEA should be used. IDEA can also help with checking procedures. An indirect way of checking procedures is to draw an inference about the effectiveness of procedures based on the results of substantive tests; e.g., it may be inferred that if there are no errors then the procedures must be working. Conversely, if there are errors then it may be inferred that procedures are being applied incorrectly and that controls are not effective. IDEA can also be used when checking computer-based controls by re-performing edit checks, matching and other computer-based procedures.

Refer to subsequent sections in this Workbook or the online Help available from the **Help** menu in IDEA to learn more about the uses of IDEA and its tests.

Volume

Classically, data analysis techniques are appropriate with a large volume of small value transactions. Generally, this is true with IDEA; therefore, the greater the volume of data the more useful IDEA will be for the investigation. If there are fewer than a few hundred records, then IDEA is unlikely to provide much benefit over manual work.

For large files, say over 100,000 records, then a calculation of the required disk size should be made (i.e., number of records multiplied by the approximate space each record takes). IDEA can be used on files with several million records, but a fast computer with a large amount of disk space will be essential.

The amount of data available for each item also makes a difference to the potential benefit of using IDEA. Items with a full history and comprehensive detail will allow a wide variety of tests. IDEA will be restrained to some simple calculations, analysis, and sampling if only the sparsest information is available.

Typically, IDEA will be useful when a client has:

- Accounts Receivable: more than 200 balances and 1,000 transactions
- Fixed Assets: more than 1,000 items
- Inventories: more than 1,000 stock lines
- Purchases: more than 2,500 transactions

Ease of Download

If there is a significant cost in using IDEA, then it is likely to be incurred when downloading the data. The data must be transferred from the host (source) computer to the computer running IDEA or onto a network location accessible by the auditor/analyst's computer. This process can be simple and only take a few minutes, or it can be more involved and requires a query or a written report. It may also require the use of equipment to convert data or to read a particular format. If you are using a bureau or agency, they may charge for their services. You may also require some technical assistance that could incur a charge for their services.

In most cases, the client supplies you the accounting data files, a printout of the data, or email messages.

Once a downloading method has been set up and used for a system or computer, subsequent downloads are usually straightforward, and the costs (if any) will be known. However, for the first time, an estimate of time and costs will need to be made.

When to Use IDEA

You can assess the likely benefits of using IDEA from the audit objectives and expected problems, as well as the volume and width of data. Additional consideration should be given to the usefulness of additional analysis over what is currently provided by the system and whether any special factors apply, such as fraud detection and investigation, and special investigations.

Costs are largely determined by the effort involved in obtaining and downloading data. There is some time in performing the IDEA tests, although this is not normally significant. It is also usual that far more "exceptional" items and queries are identified when using IDEA than with other methods, and these may require follow-up time. Alternatively, these items may be given to management for follow-up.

Using IDEA often replaces other tests and provides overall time savings. Alternatively, the reason for using IDEA may be to add to the scope, in which case the extra costs should be defined. You should balance the cost of using IDEA against the benefits.

In summary, use IDEA in audits where:

- There is a core purpose or reason
- There are a reasonable number of records
- There is a depth of information on the items
- The data transfer is technically feasible at a reasonable cost and volumes can be accommodated on computers or networks